CHAPTER - 14

TIME SERIES ANALYSIS AND FORECASTING

14.1. Introduction

Forecasting or prediction is an essential tool in any decision-making. Its uses vary from determining inventory requirements for a local shoe store to estimating annual sales of general stores. Time series analysis is one quantitative method that can be used to determine patterns in data collected over time and to project these patterns to arrive at an estimate for the future. Hence, time series analysis helps cope with uncertainly about the future.

In this chapter, we will deal with issues involved in analyzing a special type of data set, called time series data. In particular, we will be interested in analyzing the measurements through time on a particular variable. However, the statistical data can be arranged in a number of ways- according to magnitude or size, according to place of occurrence or geographical location and according to the time of occurrence or in chronological order. When the data are arranged according to the time of their occurrence, they form a time series. Thus time series is the arrangement of statistical data in chronological order of time such as hourly, daily, weekly, fortnightly, monthly or yearly. For example, monthly product, monthly sales, quarterly corporate earnings, daily closing price of share, etc. Time series are of particular importance in the field of business and economics because variables like price, wages, production, sales, profits, etc. vary from one time period to another. A time series is a set of observations taken at specified times, usually at equal intervals. According to W.Z. Harisch "A time series may be defined as a sequence of values of some variables corresponding to successive points of times". Time series analysis is used to detect patterns of change in statistical information over a regular interval of time.

Definition. A time series is a set of numerical measurements on a time-dependent variable of interest arranged over a regular interval of time.

Suppose, $y_1, y_2, \ldots\ldots\ldots, y_k$ be the values of a variable for k different times periods $t_1, t_2 \ldots t_k$ respectively, thus y is a function of t. Symbolically, $y = f(t)$.

Time series data typically possess special characteristics that necessitate statistical methods for their analysis. This type of data may be analyzed to achieve some or all of the following objectives.

14.2. Objectives of Time Series Analysis

The main objective of analyzing the time series data is to get a concrete idea about the past behaviour of data so that appropriate course of action for future can be taken. However, the objective can be pointed out as follows:

i) To identify the pattern and trends, and isolate the influencing factors or effects

ii) To apply the idea obtained from analyzing the pattern of time series data for future planning and control.

14.3. Importance of Time Series Analysis in Business Decision-Making

Time series analysis is of great importance to business executives. It is extremely useful for him/her in decision-making due to the following reasons:

i) This is the most popular and so far the effective method for business forecasting.

ii) It helps in understanding the past behaviour of economic process and in predicting the future behaviour.

iii) It helps in planning future operations.

iv) It helps in evaluating current achievement.

v) It facilitates comparison of same phenomenon over two or more periods.

14.4. Components of a Time Series

It can be easily observed that the patterns or behaviour of different time series are different. This variation is due to presence of the affects of a number of inherent components. Hence, one way of thinking about the variation in bahviour of an actual observed series is to regard it as being made up of various components. Traditionally, there are four possible components, which have been considered with the notion that any or some or all might be present in any particular series of data. These components seem to be independent of each other and seem to be influencing the overall time series data. The components are as follows:

i) Secular Trend (T_t)

ii) Cyclical components (C_t)

iii) Seasonal components (S_t)

iv) Irregular components (I_t)

14.5. Mathematical Models for Time - Series Analysis

There are two mathematical models, which are commonly used for the decomposition of a time series into different components. These are:

(i) Multiplicative model

(ii) Additive model

$$b = \frac{\sum xy}{\sum x^2} \text{ and } a = \bar{y}$$ which are simpler than the previous values.

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Example 14.7.4. Fit a trend line to the following data by Least squares method and estimate the trend component of time series data.

Year	1996	1997	1998	1999	2000	2001	2002	2003	2004	2005	2006	2007	2008
Sales Turnover (Taka in lac)	80	85	97	110	160	94	86	174	180	200	135	120	105

Solution. We have to fit the linear trend line $y = a + bx$, where y represents the sales turnover and x represents the coded year. The values of a and b are to be computed using the formula

$$b = \frac{\sum xy}{\sum x^2} \text{ and } a = \bar{y}$$

Let us construct the following table for necessary calculations of sum of squares and sum of product.

Table 14.4. Calculations of Trend values of turnover by Least Squares method.

Year	Turnover (y)	Coded period (x) Taking origin at 2002	x^2	xy	Estimated Trend values
1996	80	–6	36	–480	96.52
1997	85	–5	25	–425	101.28
1998	97	–4	16	–388	106.04
1999	110	–3	9	–330	110.80
2000	160	–2	4	–320	115.56
2001	94	–1	1	–94	120.32
2002	86	0	0	0	125.08
2003	174	1	1	174	129.84
2004	180	2	4	360	134.60
2005	200	3	9	600	139.36
2006	135	4	16	540	144.12
2007	120	5	25	600	148.88
2008	105	6	36	630	153.64
Total	$\Sigma y = 1626$	$\Sigma x = 0$	$\Sigma x^2 = 182$	$\Sigma xy = 867$	

From table 14.4, we have, $\Sigma x = 0$, $\Sigma y = 1626.00$, $\Sigma x^2 = 182$ and $\Sigma xy = 867$.

Using the formula mentioned above, we get the estimated values of b and a as $b = 4.76$ and $a = 125.08$

The fitted equation of trend line is $\hat{y} = 125.08 + 4.76\,x$

From the fitted equation, the trend values are estimated as in the following way:

For 1996, $\hat{y}_{1996} = 125.08 + 4.76\,(-6) = 96.52$

For 1997, $\hat{y}_{1997} = 125.08 + 4.76\,(-5) = 101.28$ and so on.

We can also project the trend value for the year 2010 just by putting the value of x equal to 8 which is $\hat{y}_{2010} = 125.08 + 4.76\,(-8) = 163.16$. Thus we can say that the expected sales turnover for the year 2010 is taka 163.16 lac. In this way trend value for any expected time period can be estimated.

The trend values of turnover obtained by least squares method along with original are presented in figure 14.7.

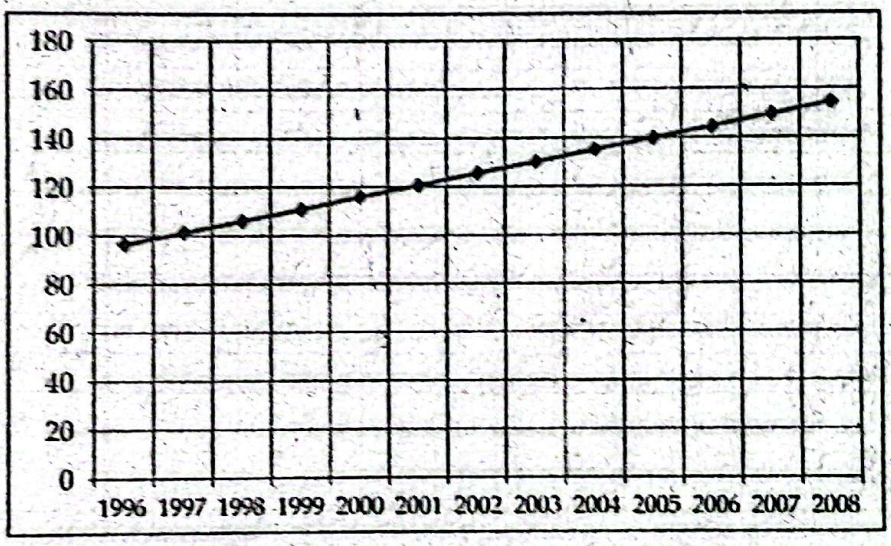

Fig. 14.7. *Least squares trend values of turnover.*

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Example 14.7.5. The following are the annual profits in thousand of taka in a certain business. Use the method of least squares to determine the trend values of profit. Also forecast the annual profits for 2009.

Year	Profit	Year	Profit
1999	60	2004	85
2000	72	2005	95
2001	75	2006	101
2002	65	2007	107
2003	80	2008	115

Solution. We have to fit the linear trend line y = a + bx, where y represents the sales turnover and x represents the coded year. Here, the number of years is 10, which is even. So, in order get the convenient values of coded year (x), we have to multiply difference of each year from mean year by 2.